Part2: TAD-Insulation and Intra-TAD Interactions

Xiaohan Xie

0. Overview

This section investigates how SARS-CoV-2 infection perturbs chromatin architecture at the level of topologically associating domains (TADs). Specifically, we reproduce three analyses from the original study:

Specifically, we aim to reproduce the following analyses in the original paper: 1. Boundary integrity: whether TAD boundary strength changes upon infection; 2. Intra-TAD interactions: whether contact frequency within TADs is altered; 3. Direct comparison of Hi-C contact matrices at a representative TAD domain.

```r
library(HiCExperiment)
```

```
Consider using the `HiContacts` package to perform advanced genomic operations
on `HiCExperiment` objects.

Read "Orchestrating Hi-C analysis with Bioconductor" online book to learn more:
https://js2264.github.io/OHCA/
```

```r
library(HiContacts)
```

```
Registered S3 methods overwritten by 'readr':
  method                    from
  as.data.frame.spec_tbl_df vroom
  as_tibble.spec_tbl_df     vroom
  format.col_spec           vroom
  print.col_spec            vroom
  print.collector           vroom
  print.date_names          vroom
```

```
  print.locale               vroom
  str.col_spec               vroom
```

```r
library(GenomicRanges)
```

```
Loading required package: stats4

Loading required package: BiocGenerics

Loading required package: generics


Attaching package: 'generics'


The following objects are masked from 'package:base':

    as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
    setequal, union


Attaching package: 'BiocGenerics'


The following object is masked from 'package:HiCExperiment':

    as.data.frame


The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs


The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
    order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
    rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
    unsplit, which.max, which.min
```

```
Loading required package: S4Vectors


Attaching package: 'S4Vectors'

The following object is masked from 'package:HiCExperiment':

    metadata<-

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Loading required package: Seqinfo
```

```r
library(InteractionSet)
```

```
Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
```

```
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians

The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians
```

```r
library(rtracklayer)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
```

```
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x lubridate::%within%() masks IRanges::%within%()
x dplyr::collapse()     masks IRanges::collapse()
x dplyr::combine()      masks Biobase::combine(), BiocGenerics::combine()
x dplyr::count()        masks matrixStats::count()
x dplyr::desc()         masks IRanges::desc()
x tidyr::expand()       masks S4Vectors::expand()
x dplyr::filter()       masks stats::filter()
x dplyr::first()        masks S4Vectors::first()
x dplyr::lag()          masks stats::lag()
x ggplot2::Position()   masks BiocGenerics::Position(), base::Position()
x purrr::reduce()       masks GenomicRanges::reduce(), IRanges::reduce()
x dplyr::rename()       masks S4Vectors::rename()
x ggplot2::resolution() masks HiCExperiment::resolution()
x lubridate::second()   masks S4Vectors::second()
x lubridate::second<-() masks S4Vectors::second<-()
x dplyr::slice()        masks IRanges::slice()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(epiwraps)
```

```
Loading required package: EnrichedHeatmap
Loading required package: grid
Loading required package: ComplexHeatmap
========================================
ComplexHeatmap version 2.28.0
Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
Github page: https://github.com/jokergoo/ComplexHeatmap
Documentation: http://jokergoo.github.io/ComplexHeatmap-reference

If you use it in published research, please cite either one:
- Gu, Z. Complex Heatmap Visualization. iMeta 2022.
- Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
    genomic data. Bioinformatics 2016.


The new InteractiveComplexHeatmap package can directly export static
complex heatmaps into an interactive Shiny app with zero effort. Have a try!

This message can be suppressed by:
  suppressPackageStartupMessages(library(ComplexHeatmap))
========================================
```

```
========================================
EnrichedHeatmap version 1.42.0
Bioconductor page: http://bioconductor.org/packages/EnrichedHeatmap/
Github page: https://github.com/jokergoo/EnrichedHeatmap
Documentation: http://bioconductor.org/packages/EnrichedHeatmap/

If you use it in published research, please cite:
Gu, Z. EnrichedHeatmap: an R/Bioconductor package for comprehensive
visualization of genomic signal associations. BMC Genomics 2018.

This message can be suppressed by:
  suppressPackageStartupMessages(library(EnrichedHeatmap))
========================================
```

```
Warning: replacing previous import 'IRanges::median' by 'stats::median' when
loading 'epiwraps'
```

```r
library(patchwork)
library(GenomicFeatures)
```

```
Loading required package: AnnotationDbi

Attaching package: 'AnnotationDbi'

The following object is masked from 'package:dplyr':

    select
```

1. Data Preparation

Using the command line tool of cooler, we merged the replicates for Mock and SARS-CoV-2-infected conditions. In accordance with the original study's methods, we processed the data at 10kb resolution. Because the paper does not specify a normalisation procedure in detail, we applied cooler's built-in matrix-balancing algorithm to correct for coverage biases and generated balanced contact scores, which were used throughout the downstream analysis. Lastly, we performed matrix coarsening to generate multi-resolution data, as this was needed for the calculation of insulation scores.

We limited our analysis to chromosome 14 to reduce the computational complexity.

```
cooler merge merged_mock_HiC.mcool mock_HiC_rep1.mcool::/resolutions/10000 mock_HiC_rep2.mcoo
cooler merge merged_cov_HiC.mcool cov_HiC_rep1.mcool::/resolutions/10000 cov_HiC_rep2.mcool:
cooler balance merged_mock_HiC.mcool
cooler balance merged_cov_HiC.mcool
cooler zoomify -r 10000,200000 merged_mock_HiC.mcool -o merged_mock_HiC_multi.mcool
cooler zoomify -r 10000,200000 merged_cov_HiC.mcool -o merged_cov_HiC_multi.mcool
```

```
hic_mock_merged <- HiCExperiment("./GSE179184_RAW/merged_mock_HiC_multi.mcool", focus = "chr1
hic_cov_merged <- HiCExperiment("./GSE179184_RAW/merged_cov_HiC_multi.mcool", focus = "chr14"
length(hic_mock_merged)
```

```
[1] 3829954
```

```
length(hic_cov_merged)
```

```
[1] 3728822
```

```
summary(interactions(hic_mock_merged)$balanced)
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.       NAs
5.846e-05 2.257e-04 3.452e-04 1.193e-03 6.174e-04 2.909e-01     43969
```

```
summary(interactions(hic_cov_merged)$balanced)
```

```
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.       NAs
5.072e-05 2.495e-04 3.922e-04 9.968e-04 6.773e-04 1.201e-01     57439
```

Mock sample has slightly more interaction counts than the CoV-infected group. Both samples show highly skewed distribution, with the majority of interactions having very low balanced scores, which is expected for Hi-C data.

2. Insulation Scores and Boundary strengths

According to methods decribed in the original paper, we calculated the insulation scores using getDiamondInsulation, setting a 200kb sliding window. TAD boundaries were then called by finding the local minima in the insulation profile with a threshold of 0.5.

```r
hic_mock <- getDiamondInsulation(hic_mock_merged, window_size = 200000)
```

```
Going through preflight checklist...

Scan each window and compute diamond insulation score...

Annotating diamond score prominence for each window...
```

```r
hic_cov <- getDiamondInsulation(hic_cov_merged, window_size = 200000)
```

```
Going through preflight checklist...

Scan each window and compute diamond insulation score...

Annotating diamond score prominence for each window...
```

```r
hic_mock <- getBorders(hic_mock, strong_threshold = 0.5)
hic_cov <- getBorders(hic_cov, strong_threshold = 0.5)
bounds_mock <- topologicalFeatures(hic_mock, 'borders')
bounds_cov <- topologicalFeatures(hic_cov, 'borders')
```

To allow a direct, position-matched comparison between conditions, we pooled the boundaries called in the Mock and infected samples into a single consensus set, as done in the original study.

```r
merged_bounds <- GenomicRanges::reduce(c(bounds_mock, bounds_cov))
merged_bounds <- merged_bounds[seqnames(merged_bounds) == "chr14"]
head(merged_bounds)
```

```
GRanges object with 6 ranges and 0 metadata columns:
      seqnames            ranges strand
         <Rle>         <IRanges>  <Rle>
  [1]    chr14 19250001-19260000      *
  [2]    chr14 19460001-19470000      *
  [3]    chr14 19790001-19800000      *
  [4]    chr14 20000001-20010000      *
  [5]    chr14 20690001-20700000      *
  [6]    chr14 20740001-20750000      *
  -------
  seqinfo: 25 sequences from an unspecified genome
```

```r
ins_track_mock <- metadata(hic_mock)$insulation
ins_track_cov <- metadata(hic_cov)$insulation
```

We then used the **epiwraps** package to visualise insulation-score profiles centred on the consensus boundaries, extending 500 kb up- and downstream of each boundary.

```r
ins_track_mock$score <- ins_track_mock$insulation
ins_track_cov$score  <- ins_track_cov$insulation
tracks <- list(Mock = ins_track_mock, SARS_CoV_2 = ins_track_cov)
ins <- signal2Matrix(tracks, regions = merged_bounds, extend = 500000, w = 10000)
```

```
Computing signal from GRanges 'Mock'...

Computing signal from GRanges 'SARS_CoV_2'...
```

```r
plotEnrichedHeatmaps(ins, scale_title ="Insulation Scores", use_raster = FALSE)
```

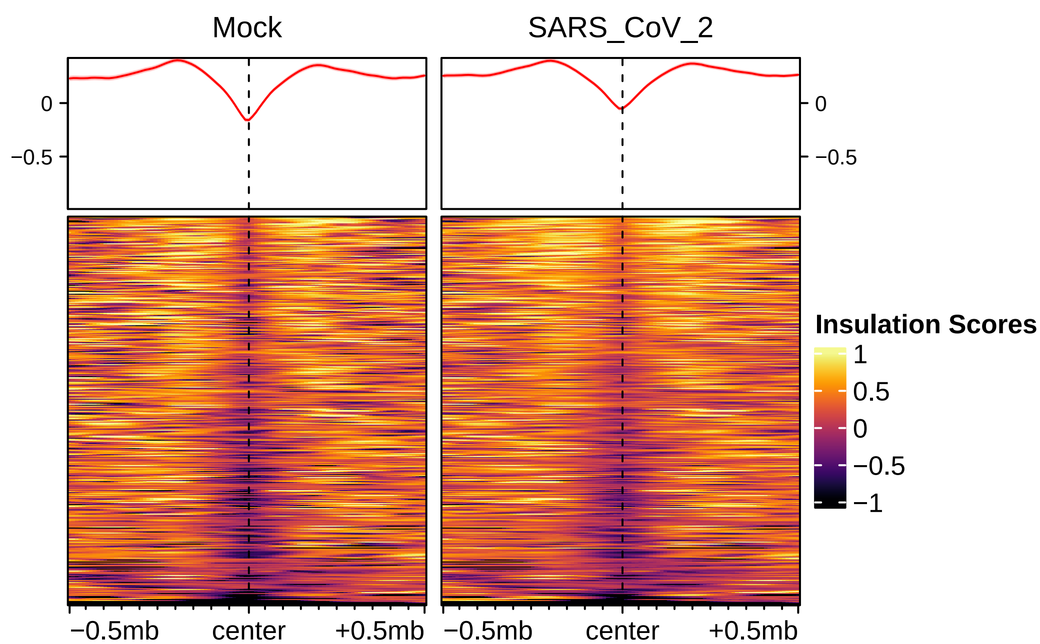

Figure 1: Insulation scores calculated TAD boundaries in both conditions.

We can see that although CoV-infected group shows a slightly higher insulation score at the consensus boundaries, this effect is rather mild, consistent with the original study (Fig.4d). This indicates that TAD boundary strength is largely preserved upon infection.

3. TADs and Intra-TAD Interaction

We then further obtained the TADs, defined as genomic intervals nested between the previously identified consensus boundaries.

```
# gaps between merged boundaries are TADs
merged_tads <- gaps(merged_bounds)
merged_tads <- merged_tads[strand(merged_tads) == "*"]
merged_tads <- merged_tads[seqnames(merged_tads) == "chr14"]
# filter out extremely small fragments (as we have resolution of 10000, use threshold 20000)
merged_tads <- merged_tads[width(merged_tads) >= 20000]
# get the GInteractions objects
gi_mock <- interactions(hic_mock)
gi_cov <- interactions(hic_cov)
# filter out very short distant interactions (< 15 kb)
gi_mock_sub <- gi_mock[pairdist(gi_mock) >= 15000]
gi_cov_sub <- gi_cov[pairdist(gi_cov) >= 15000]
```

To quantify the intra-TAD interactions, we defined a helper function to isolate interactions that occur strictly within the same TAD domain. We then computed the mean balanced contact score per TAD for each condition.

```
# func to map interactions inside TADs
get_intra_tad_means <- function(gi_filtered, tads) {

  hits1 <- findOverlaps(anchors(gi_filtered, type="first"), tads)
  hits2 <- findOverlaps(anchors(gi_filtered, type="second"), tads)

  # keep only interactions where both anchors found in the same TAD
  common_hits <- inner_join(
    as.data.frame(hits1),
    as.data.frame(hits2),
    by = "queryHits", suffix = c("1", "2")
  ) %>% filter(subjectHits1 == subjectHits2)

  # aggregate mean contact score per TAD

  df <- data.frame(
    tad_idx = common_hits$subjectHits1,
    contact_val = gi_filtered$balanced[common_hits$queryHits]
  )
```

```r
  df_mean <- df %>%
    group_by(tad_idx) %>%
    summarize(mean_contact = mean(contact_val, na.rm = TRUE))

  return(df_mean)
}

mock_means <- get_intra_tad_means(gi_mock_sub, merged_tads)
cov_means <- get_intra_tad_means(gi_cov_sub, merged_tads)
```

To compare conditions, we then computed, for each TAD, the log2 fold-change of the mean contact score (SARS-CoV-2 relative to Mock).

```r
tad_metrics <- data.frame(tad_id = 1:length(merged_tads)) %>%
  left_join(mock_means, by = c("tad_id" = "tad_idx")) %>%
  rename(Mock_Mean = mean_contact) %>%
  left_join(cov_means, by = c("tad_id" = "tad_idx")) %>%
  rename(Cov_Mean = mean_contact) %>%
  mutate(
    log2FC = log2((Cov_Mean) / (Mock_Mean))
  )

# save it as a col
mcols(merged_tads) <- tad_metrics
```

We then visualised the distribution of these log2FCs.

```r
# Fig.4c: boxplot of intra-TAD interaction changes
ggplot(tad_metrics, aes(x = "Infection", y = log2FC)) +
  geom_violin(fill = "#E74C3C", width = 0.4) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "black") +
  labs(
    x = "",
    y = expression(log[2]~"Fold Change (SARS-CoV-2 / Mock)")
  ) +
  theme_classic(base_size = 14)
```

```
Warning: Removed 35 rows containing non-finite outside the scale range
(`stat_ydensity()`).
```

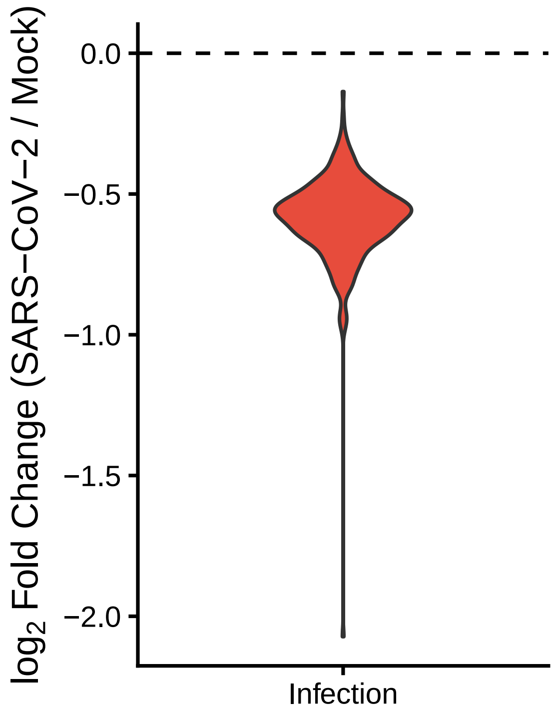

Figure 2: log2FCs of intra-TAD Hi-C interactions.

It can be seen from the plot that SARS-CoV-2 infection reduces the intra-TAD contacts, which aligns with the findings of the paper.

4. Hi-C Matrices at a Specific TAD Domain

Following the original study (Fig.4a), we defined a specific TAD domain of interest on chromosome 14 and visualised Hi-C contacts at this region. We proceeded with bin size of 10kb, unlike in the paper, where the authors switched to a resolution of 5kb for this analysis. We calculated a differential contact matrix for this TAD region, which we visualised together with the contact matrices of both Mock and CoV groups.

```r
# the target genomic coordinates
target_chr <- "chr14"
target_start <- 71400000
target_end   <- 72600000
target_region <- paste0(target_chr, ":", target_start, "-", target_end)
target_gr <- GRanges(seqnames = target_chr, ranges = IRanges(target_start, target_end))
```

```r
hic_mock_reg <- refocus(hic_mock, target_region)
hic_cov_reg <- refocus(hic_cov,  target_region)

hic_div      <- divide(hic_cov_reg, by = hic_mock_reg, use.scores = "balanced")
```

```r
# mock
p_matrix_mock <- plotMatrix(hic_mock_reg, use.scores = "balanced", scale = "log10", caption =
  labs(title = "Mock", fill = "log10 Score", y = paste0(target_chr, ": 71,400,000-72,600,000"
  theme(axis.text = element_blank(), axis.ticks = element_blank())

# cov
p_matrix_cov <- plotMatrix(hic_cov_reg, use.scores = "balanced", scale = "log10", caption = F
  labs(title = "SARS-CoV-2", fill = "log10 Score") +
  theme(axis.title.y = element_blank(), axis.text = element_blank(), axis.ticks = element_bla

# log2 FC
p_matrix_l2fc <- plotMatrix(
  hic_div,
  use.scores = "balanced.l2fc",
  scale = "linear",
  cmap = bgrColors(), caption = FALSE,
  limits = c(-2, 2)
) +
  labs(title = "SARS-CoV-2/mock", fill = "log2 FC") +
  theme(axis.title.y = element_blank(), axis.text = element_blank(), axis.ticks = element_bla

matrix_panels <- p_matrix_mock + p_matrix_cov + p_matrix_l2fc +
  plot_layout(ncol = 3)

matrix_panels
```

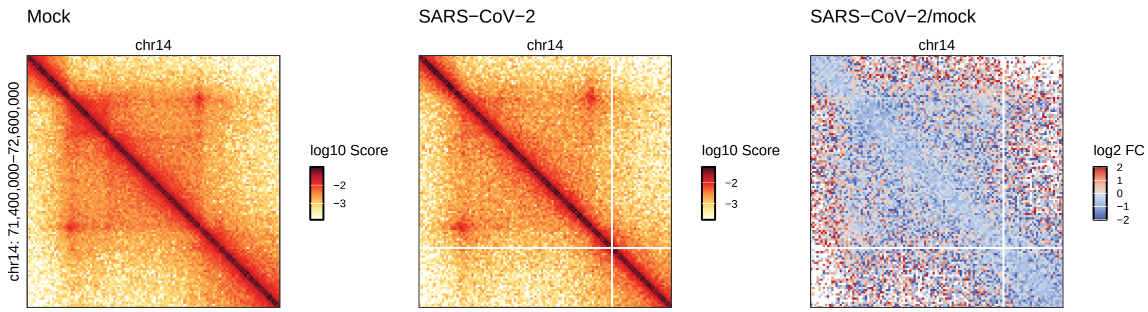

Figure 3: Hi-C matrices (bin size = 10 kb) at the indicated region (hg19) and log2FC after infection.

The patterns here indicate that while intra-TAD interactions tend to be reduced by infection of SARS-CoV-2, contacts outside of TADs (off-diagonal) are largely unaltered or even increased. This was demonstrated more comprehensively by the Aggregate Domain Analysis in the original paper (Fig.4b). Looking at the contact matrices of the two groups (Mock and CoV), we can also see that TAD-boundaries are not noticeably perturbed by infection, corroborating the results of the insulation score analysis.

5. Discussion

This part of the analysis shows that SARS-CoV-2 infection selectively weakens intra-TAD interactions, while leaving TAD boundaries and inter-TAD contacts largely intact.

It should be noted that our pipeline does not exactly replicate the original methods, since the paper did not fully specify several parameter choices. We were also unable to reproduce the Aggregate Domain Analysis due to a persistent function error that we could not resolve.

Despite these methodological differences, the trends we recovered from this part align well with the conclusions of the original paper.